Fiscal Year:	FY 2006	**Task Last Updated:**	FY 11/28/2006
PI Name:	Chapes, Stephen K. Ph.D.		
Project Title:	Differentiation of Bone Marrow Macrophages in Space		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	None		
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Zip Code:	66506-4901	**Congressional District:**	2
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	96-HEDS-04
Start Date:	12/28/1998	**End Date:**	03/31/2008
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	Shuttle/ISS		
Flight Assignment:	STS115 NOTE: Received NCE to 3/31/2008 per A. Chu/ARC (7/07) NOTE: Received NCE through 9/30/2007, per A. Chu (9/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG2-1274		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Space flight and/or microgravity have dramatic effects on macrophage and other cells. For example, murine bone marrow macrophages respond to microgravity in a period of about 8 seconds. These early responses to microgravity are associated with observations that bone marrow stem cell differentiation into mature macrophages is inhibited during space flight. The specific aim of this project is to determine mechanisms for the disrupted macrophage differentiation during space flight. I propose to test two hypotheses to address this specific aim: (1) Macrophage differentiation is affected because the receptor for CSF-1, c-fms is down regulated on macrophage precursors, and (2) Space flight disrupts c-fos-dependent processes involved in macrophage differentiation.We will use SHOT Inc. hardware, the Cellcult unit and bioreactor to grow primary bone marrow in the presence of macrophage colony stimulating factor. All experimental preparations will be done at the PI's home institution or at Morehouse University in Atlanta. Therefore, this project meets the requirement for needing limited crew training and involvement as described in NRA 96-HEDS-04. Moreover, this project has a flight history and a sound experimental rationale to justify its execution.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Space flight offers a unique environment that impacts many physiological systems. Amongst these systems, bone metabolism is known to be significantly altered by the skeletal unloading of space flight. Hematopoiesis is the process which provides the body with mature, functioning cells necessary for oxygenation, clotting and immune functions. The bone, specifically, the bone marrow, is the site for hematopoiesis.Moreover, changes in bone impact all compartments therein, including the marrow. Indeed, analysis of animals from rats to rhesus moneky to humans shows that hematopoiesis can be affected by space flights. Although direct examination of primary hematopoiesis is not an aim of the proposed project, this experiment will allow us to dissect a very complex system. We will obtain valuable information about what mechanisms involved in hematopoiesis are susceptible to disruption by skeletal unloading. This understanding will allow for more effective countermeasure planning. On Earth, skeletal unloading occurs to individuals that are bedridden for long periods of time, wheel-chair bound, or lose the ability to stand for long periods of time. These prolonged periods of skeletal unloading also induce changes in bone that parallel those seen in space flight. In addition, individuals with bone disorders, such as osteoporosis, are also subject to alterations in hematopoiesis. This is best exemplified by the osteopetrotic mouse, which has diminished numbers of macrophages and brittle bones. Therefore, data obtained from this study will also have direct extrapolations to individuals on Earth.
Task Progress:	In order to gain a better understanding of macrophage function during space flight, we have initiated a series of ground-based studies to understand how macrophages change in different microenvironments. Macrophages are versatile cells, able to take on various phenotypes to fulfill their vital and diverse roles in host defense. The response of the activated macrophages depends on the environment in which they reside. Knowing how macrophages will respond is important when using them for cell therapy or in understanding how to change their responses to lessen the severity of a disease. This study provides insight into the cellular in vivo changes of an established macrophage cell line after adoptive transfer. To study the changes of the C2D macrophage cell line in vivo we investigated the trafficking patterns and cellular changes after intraperitoneal adoptive transfer. In vitro, C2D macrophages reside early in the macrophage cell lineage. C2D macrophages collected from the peritoneal cavity showed increased levels of CD11b and F4/80 on their surface as compared to in vitro C2D macrophages. There were also two small populations that expressed either CD11c, a dendritic cell marker, or Gr-1, a granulocyte marker. We found that approximately 25% of the adoptively transferred C2D macrophages remained in the peritoneal cavity while others were detected to be integrated into in the intestine and spleen by seven day post transfer. These data indicates that C2D macrophages are able to respond to the host environment making them a very unique tool for studying the spaceflight and altered gravity vector environements and may be useful for ground-based immunotherapies and restoration of immune responses. These experiments are now being compiled for a manuscript to be published in 2007.
Bibliography Type:	Description: (Last Updated: 09/15/2021)
Abstracts for Journals and Proceedings	Potts BE, Chapes SK. "In vivo responses and trafficking patterns of C2D macrophages after intraperitoneal adoptive transfer. " Presented at the 39th Annual Meeting of the Society for Leukocyte Biology, San Antonio, TX, November 9-11, 2006. J. Leukocyte Biol. 2006 S:84. , Oct-2006
Abstracts for Journals and Proceedings	Potts BE, Snyder LL, Chapes SK. "In vitro migration and tissue specific responses of C2D macrophages to adipocytes. " Presented at the National IDeA Symposium of Biomedical Research Excellence (NISBRE), Washington, D.C., July 20-22, 2006. National IDeA Symposium of Biomedical Research Excellence (NISBRE), Abstracts, p. C283, 2006. , Jul-2006
Abstracts for Journals and Proceedings	Snyder LL, Potts BE, Chapes SK. "In vitro migration and tissue specific responses of C2D macrophages to adipocytes. " Presented at the 39th Annual Meeting of the Society for Leukocyte Biology, San Antonio, TX, November 9-11, 2006. J. Leukocyte Biol. 2006;Supplement addendum:10. , Oct-2006